FINAL YEAR PROJECT · AI ENGINEERING · CVPR CONFERENCE STYLE

ArSignTutor: Interactive Real-Time Arabic Sign Language Learning System

Using Deep Learning & Leakage-Aware Evaluation

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KEYWORDS: MobileNetV2 · Transfer Learning · dHash Clustering · Data Leakage · Real-Time · Arabic NLP · PyTorch · Streamlit

INPUT 224×224 — FEATURES Conv — MBv2 Blocks — FC 1280 — 31 CLASSES

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31 Arabic Alphabet Classes

ACCURACY 94.33%

SECTION 01

WHY ARABIC SIGN LANGUAGE?

Community Need
7M+ Arabic-speaking deaf individuals with very few dedicated learning tools or interactive systems.

Research Gap
ASL dominates research — Arabic Sign Language remains severely underexplored despite community size.

Tool Deficit
Existing systems are static classifiers only. No interactive tutoring or real-time feedback for ArSL.

Evaluation Crisis
High benchmark scores hide leakage. Honest evaluation requires cross-dataset and cluster-based testing.

SECTION 04

PREPROCESSING & AUGMENTATION

RandomRotation
±15°

RandomAffine
Translate 8%, Scale 90-110%

ColorJitter
Brightness, Contrast, Saturation

RandomErasing
p=0.25, scale 2-12%

Normalization
ImageNet μ/σ (0.485/0.229...)

No Flip
Mirroring changes sign meaning

Applied to training only. Validation/test: resize + normalize only.

SECTION 06

TRAINING CONFIGURATION

Optimizer	AdamW
Loss	CrossEntropy + ε=0.10
Batch Size	32
Epochs	Head: 5 | Fine-tune: ≤25
LR	1e-3 → 1e-4
Scheduler	ReduceLROnPlateau
Early Stop	Patience = 6
Precision	AMP Mixed
Device	Kaggle Tesla T4
Init	ImageNet → ArASL → RGB

SECTION 02

DATASETS

ArASL Dataset

54,049	32	40+
Images	Classes	Contributors

⚠ Near-duplicate frames → random split causes data leakage

RGB Arabic Alphabets Dataset (Independent OOD)

7,856	31	0
Images	Classes	Leaky Clusters

✓ Clean cluster-based split: 5,509 train | 1,182 val | 1,165 test

SECTION 03

METHODOLOGY PIPELINE

Dataset Analysis → dHash Cluster → Clean Split → Augment +Prep → MBv2 Phase 1 → Fine-tune Phase 2 → Eval +Deploy

Phase 1 — Head Training

Backbone	FROZEN ✗
Classifier	TRAINING ✓
LR	1e-3
Epochs	5

Phase 2 — Full Fine-Tuning

Backbone	UNFROZEN ✓
Classifier	TRAINING ✓
LR	1e-4
Epochs	≤25 + ES

SECTION 05

MODEL ARCHITECTURE — MobileNetV2

Input 224×224 → Conv BN ReLU6 → 19×Inverted Residual → Conv BN ReLU6 → Adaptive AvgPool → FC → 31 Classes

1,280	3.4M	<50ms	31
Feature Dims	Parameters	Inference	Classes

Init: ImageNet pretrained → ArASL fine-tuned → RGB domain-adapted

SECTION 12

MAIN CONTRIBUTIONS

- Developed ArSignTutor educational interactive platform for Arabic Sign Language
- Applied MobileNetV2 transfer learning with two-phase fine-tuning strategy
- Introduced leakage-aware dHash cluster-based evaluation protocol
- Performed cross-dataset validation exposing 99%→48.78% generalization gap
- ✅ Achieved 94.33% clean test accuracy — scientifically defensible result

SECTION 07

EXPERIMENTAL RESULTS

94.33%	94.59%	94.33%	94.36%
TEST ACCURACY	PRECISION	RECALL	F1-SCORE

SECTION 08

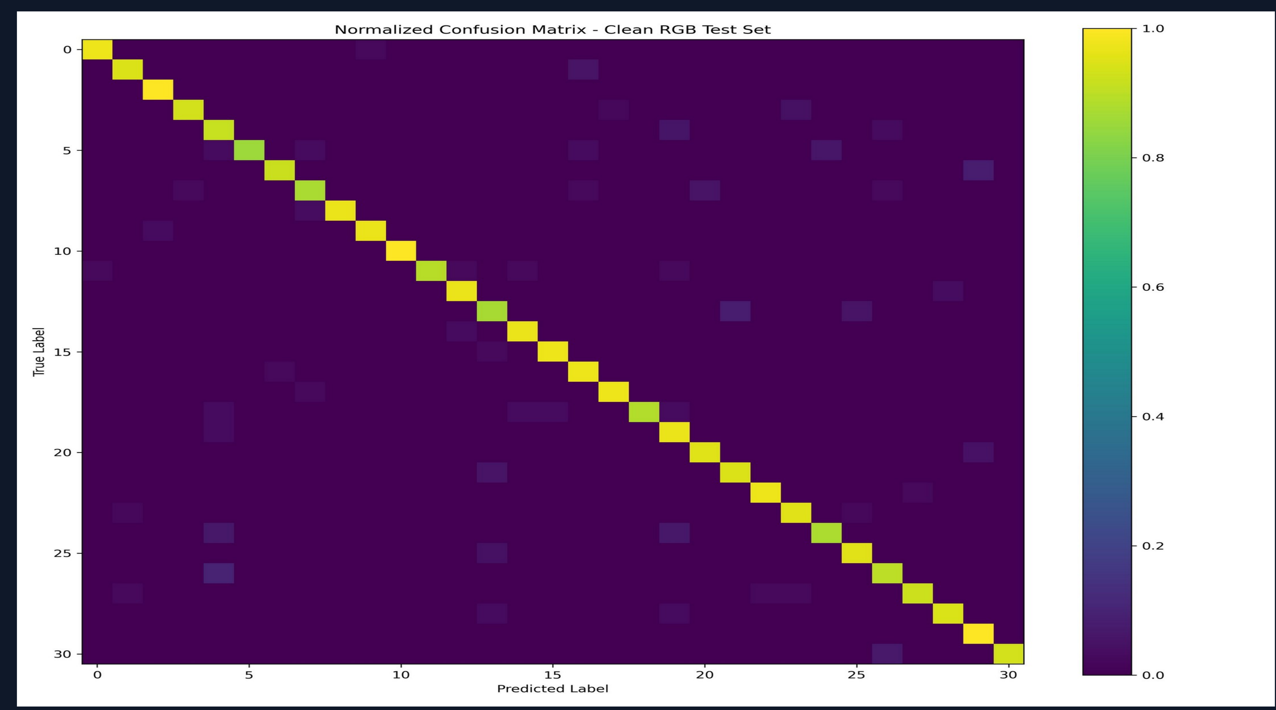
GENERALIZATION ANALYSIS

Original Model	Random (image)	ArASL	99.26%	⚠ LEAKY
Sanity Check	—	ArASL	98.75%	PIPELINE OK
Cross-Dataset OOD	ArASL → RGB	RGB 31cls	48.78%	↓ GAP
Clean Retrain ✓	Cluster (dHash)	Held-out RGB	94.33%	✓ HONEST

48.78% OOD drop = leakage + domain shift + class mapping. Clean 94.33% is the honest result.

SECTION 09

CONFUSION MATRICES



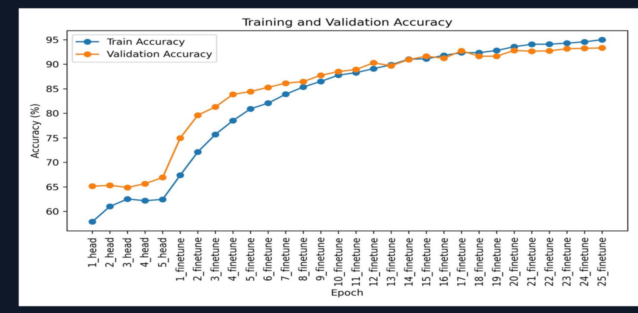
SECTION 13

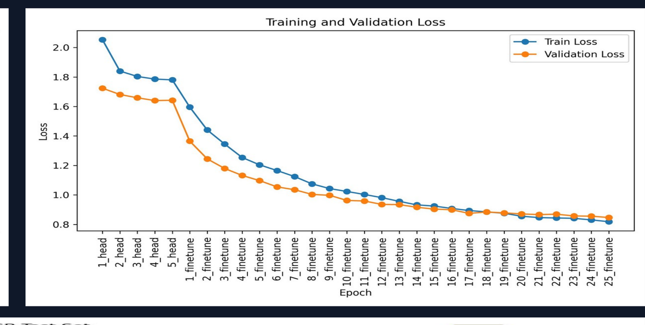
DEPLOYMENT

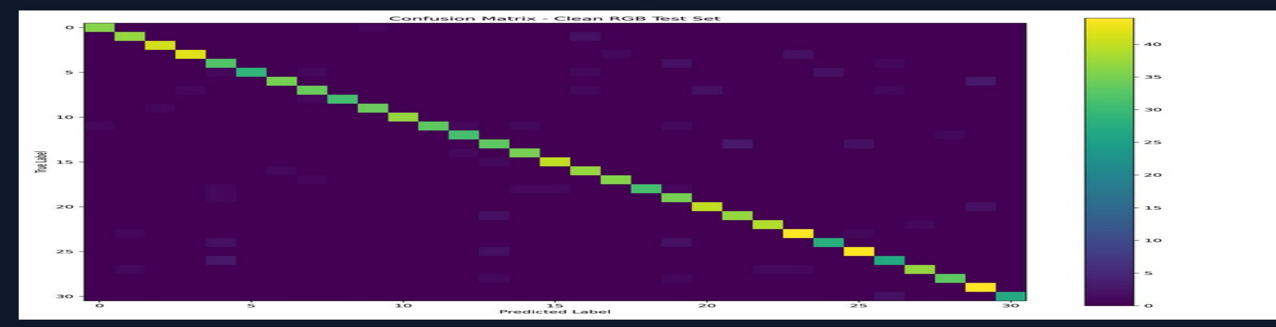
Framework	Streamlit
Model	MobileNetV2
Backend	PyTorch
Inference	Real-Time <50ms
Platform	Desktop Prototype
Classes	31 Arabic Letters

SECTION 10

TRAINING CURVES



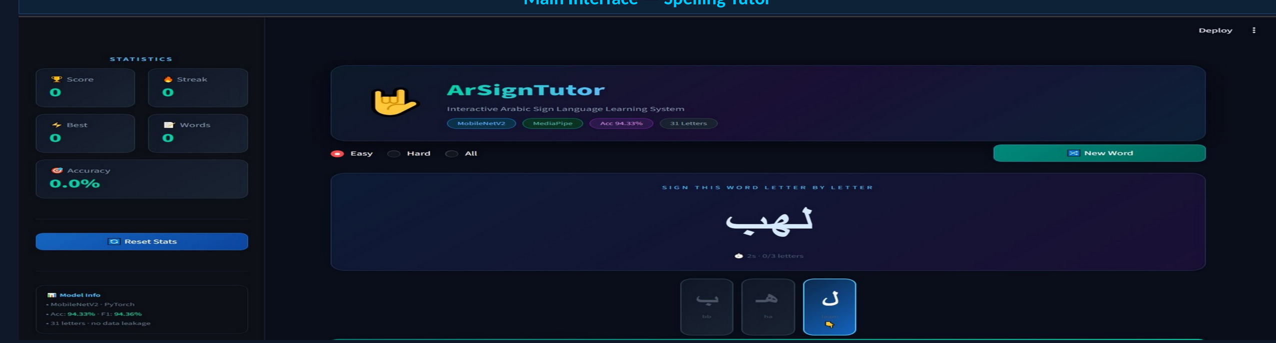




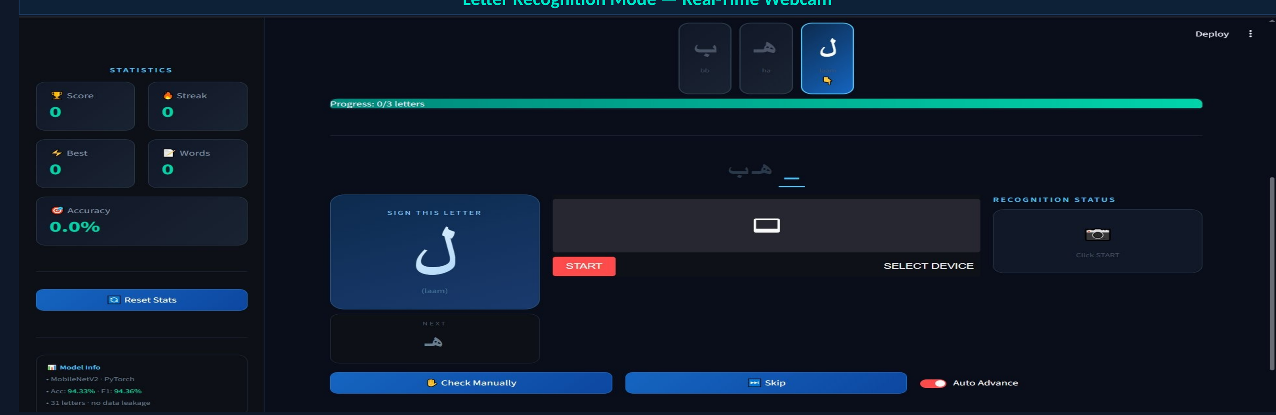
SECTION 14

SYSTEM DEMONSTRATION

Main Interface — Spelling Tutor



Letter Recognition Mode — Real-Time Webcam



SECTION 11

LIMITATIONS & FUTURE WORK

Static Signs Only
Dynamic word-level signs need video/temporal models

31 of 32 Letters
yaa excluded — no RGB samples available

Single-Frame
Temporal smoothing only partially fixes jitter

SECTION 15

CONCLUSION

ArSignTutor demonstrates that Arabic Sign Language learning can be effectively supported through an interactive deep learning system.

After removing data leakage via dHash cluster-based evaluation and domain adaptation to RGB data, the final model achieved 94.33% clean test accuracy — a scientifically honest and reproducible result.

The key lesson: a near-perfect benchmark score is not always a good model. Checking the evaluation protocol matters as much as the architecture.

- 🎬 Video temporal models for word-level signs
- ✂️ MediaPipe crop retraining
- 👁 Grad-CAM visualization
- 📱 Mobile deployment (Android/iOS)